Introduction to Network DevicesTheory

1.Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

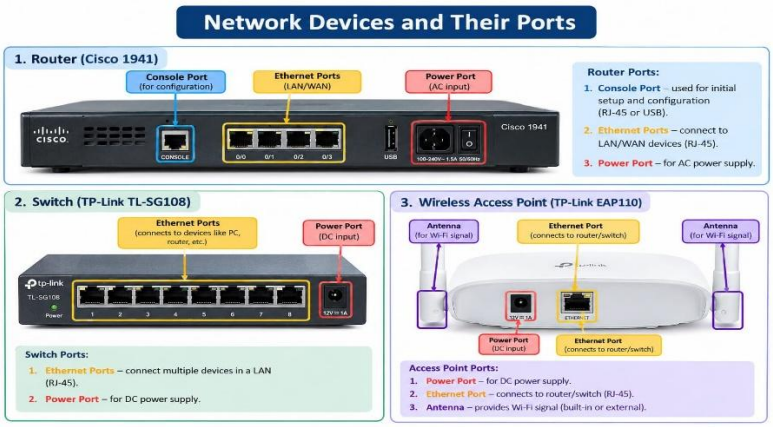
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Main Functions and Typical Ports of Network Devices

Network Device	Main Function	Typical Ports
Switch	Connects multiple wired devices within a local network (LAN).	Ethernet ports (RJ-45)
Router	Connects the home network to the internet and routes data between networks.	WAN port, LAN ports (RJ-45)
Hub	Connects multiple devices and sends incoming data to all connected ports.	Ethernet ports (RJ-45)
Firewall	Protects the network by blocking unauthorized access and filtering traffic.	Ethernet ports (RJ-45); logical TCP/UDP ports such as 80 and 443
Access Point (AP)	Allows wireless devices like smartphones and laptops to connect to the network through Wi-Fi.	Ethernet port (RJ-45), wireless radio interface

2.Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

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3.Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

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App: Instagram

1. Access Point: Connects my mobile phone to the home Wi-Fi network so I can use Instagram.
2. Router: Transfers data between my home network and the internet to access Instagram servers.
3. Switch: Connects multiple wired devices in the local network and forwards data to the correct device.
4. Firewall: Filters network traffic and blocks unauthorized access to help protect Instagram's servers.

**4.Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.**

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One possible reason for slow college Wi-Fi is that the access point cannot handle too many students connected at the same time. This causes slow speeds and frequent disconnections. Upgrading to a high-capacity Wi-Fi 6 access point with better coverage can support more devices and improve connection speed and stability.